



Building an AI Learning Buddy

A Step-by-Step Guide to Building and Deploying a Web App with Streamlit, Google Colab, ngrok, and Google Gemini

1. What We Are Building

In this session, you will build a small AI-powered web application called the AI Learning Buddy. A user types in any topic, chooses what they want (an explanation, a real-life example and a quiz etc.), and the app uses Google's Gemini AI model to generate the answer instantly in the browser.

You will use four tools together, and it is important to understand what each one does:

- Streamlit — a Python library that turns a plain script into a web page with text boxes, buttons, and dropdowns, without needing HTML or JavaScript.
- Google Colab — a free, cloud-based notebook where we write and run our Python code (no installation needed on your laptop).
- Gemini API (google-generativeai) — Google's AI model that generates the explanations, examples, and quizzes.
- ngrok — a tool that creates a public internet link to a program running privately inside Colab, so anyone with the link can open your web page.

2. Before You Start

- A Google account (to use Google Colab).
- A Gemini API key — get one for free at aistudio.google.com/app/apikey.
- A free ngrok account and auth token — sign up at ngrok.com and copy your token from the ngrok dashboard.

Important — Keep Your Keys Private

Your Gemini API key and ngrok auth token are personal and private, like a password.

Never type your real key directly into a notebook you plan to share, screenshot, upload, or submit as an assignment.

Anyone who sees your key can use your account and your free quota. If you ever paste a key into a notebook and share it, go back and regenerate (reset) that key immediately.

In this guide, keys are shown as placeholders — you will paste in your own private key when you build the app yourself.

3. Step 1 — Open Google Colab and Install Libraries

Go to colab.research.google.com and create a New Notebook. In the first code cell, install the three Python libraries the project needs: Streamlit (for the web page), pyngrok (to expose it publicly), and google-generativeai (to talk to Gemini).

```
!pip install -q streamlit pyngrok google-generativeai
```

Run this cell (Shift + Enter). The -q flag just keeps the output quiet/clean.

4. Step 2 — Write the App Code to a File

Streamlit needs your app's code saved as an actual .py file on the Colab machine — it cannot run code that is only sitting inside a notebook cell. Colab's %%writefile magic command solves this: everything typed below that line in the cell is saved into a file named app.py, instead of being executed immediately.

Paste this into a new cell:

```
%%writefile app.py

import streamlit as st
import google.generativeai as genai

# Configure Gemini API
genai.configure(api_key="YOUR_GEMINI_API_KEY_HERE")

model = genai.GenerativeModel("gemini-2.5-flash")

st.set_page_config(page_title="AI Learning Buddy", page_icon="🧠")

st.title("🧠 AI Learning Buddy")

topic = st.text_input("Enter a Topic")

option = st.selectbox(
    "Choose Activity",
    [
        "Explain Concept",
        "Real-Life Example",
        "Generate Quiz",
        "Ask Anything"
    ]
)

if st.button("Generate"):

    if topic == "":
        st.warning("Please enter a topic.")
    else:

        if option == "Explain Concept":
            prompt = f"Explain {topic} in simple language for a beginner."

        elif option == "Real-Life Example":
            prompt = f"Give one simple real-life example of {topic}."

        elif option == "Generate Quiz":
            prompt = f"Create 5 MCQs on {topic} with answers."

        else:
            prompt = topic

        response = model.generate_content(prompt)

        st.write(response.text)
```

Remember

Replace "YOUR_GEMINI_API_KEY_HERE" with your own private Gemini API key before running this cell. The %%writefile line must be the very first line of the cell.

5. Understanding the App Code, Line by Line

5.1 Setup

- `import streamlit as st` — loads Streamlit and gives it the short nickname `st`, used for every UI element.
- `import google.generativeai as genai` — loads Google's Gemini SDK.
- `genai.configure(api_key=...)` — connects the app to your Gemini account using your API key.
- `model = genai.GenerativeModel("gemini-2.5-flash")` — selects which Gemini model will generate the responses.

5.2 Page and Layout

- `st.set_page_config(...)` — sets the browser tab title and icon.
- `st.title(...)` — displays the big heading at the top of the page.
- `st.text_input(...)` — creates the box where the student types a topic.
- `st.selectbox(...)` — creates the dropdown menu with the four activity choices.

5.3 App Logic

- `st.button("Generate")` — the code inside this if-block only runs when the button is clicked.
- `if topic == "":` — checks the topic box isn't empty, and shows a warning if it is.
- The `if / elif / else` chain builds a different prompt (instruction) for Gemini depending on which activity was chosen.
- `model.generate_content(prompt)` — sends the prompt to Gemini and gets back an AI-generated response.
- `st.write(response.text)` — displays that response on the web page.

6. Step 3 — Run the Streamlit App in the Background

Colab runs one thing at a time by default, so we start Streamlit as a background process using `nohup` and `&`. This lets the server keep running while we move on to the next cell.

```
!nohup streamlit run app.py --server.port 8501 &
```

Give the server a few seconds to fully start before checking on it:

```
import time
time.sleep(10)
```

You can peek at the server's startup log to confirm it's running correctly (look for a line mentioning "Local URL" or "Network URL"):

```
!cat nohup.out
```

7. Step 4 — Expose the App with ngrok

Right now, the app is only running privately inside the Colab machine — nobody outside can open it. ngrok creates a secure tunnel that gives it a real, public web address.

4a. Install ngrok's Python package (if not already installed):

```
!pip install -q pyngrok
```

4b. Connect to your ngrok account and open the tunnel:

```
from pyngrok import ngrok

ngrok.set_auth_token("YOUR_NGROK_AUTH_TOKEN_HERE")

public_url = ngrok.connect(8501)

print(public_url)
```

- `ngrok.set_auth_token(...)` — logs the tunnel into your personal ngrok account using your token.
- `ngrok.connect(8501)` — opens a tunnel to port 8501, the same port Streamlit is running on.
- `print(public_url)` — prints the public link. Click it (or copy it into a new browser tab) to open your live AI Learning Buddy app.

Tip

The free ngrok link changes every time you rerun `ngrok.connect()`, so always use the newest link printed. If the printed link doesn't open, re-check that the Streamlit cell in Step 3 is still running.

8. Step 5 — Try Out the App

1. Open the ngrok link printed in Step 4.
2. Type any topic into the "Enter a Topic" box, e.g. Photosynthesis.
3. Choose an activity from the dropdown: Explain Concept, Real-Life Example, Generate Quiz, or Ask Anything.
4. Click Generate and watch Gemini's response appear below the button.
5. Try the same topic with a different activity to see how the response changes.

Congratulations — you've now built and deployed a real AI-powered web app entirely from a browser, without installing anything on your own computer!

9. Troubleshooting

- "Address already in use" when starting Streamlit — another Streamlit process is already running on that port. Restart the Colab runtime (Runtime → Restart runtime) and run the cells again in order.
- ngrok link shows an error page — the Streamlit app may not have finished starting; wait a few more seconds and refresh, or re-check nohup.out for errors.
- "Invalid API key" error in the app — double-check the Gemini key was pasted correctly, with no extra spaces, inside the quotes.
- Nothing happens when clicking Generate — make sure the Topic box isn't empty; the app shows a warning message in that case.
- Colab session disconnected — Colab free tier disconnects after a period of inactivity; simply rerun all cells from the top in order.